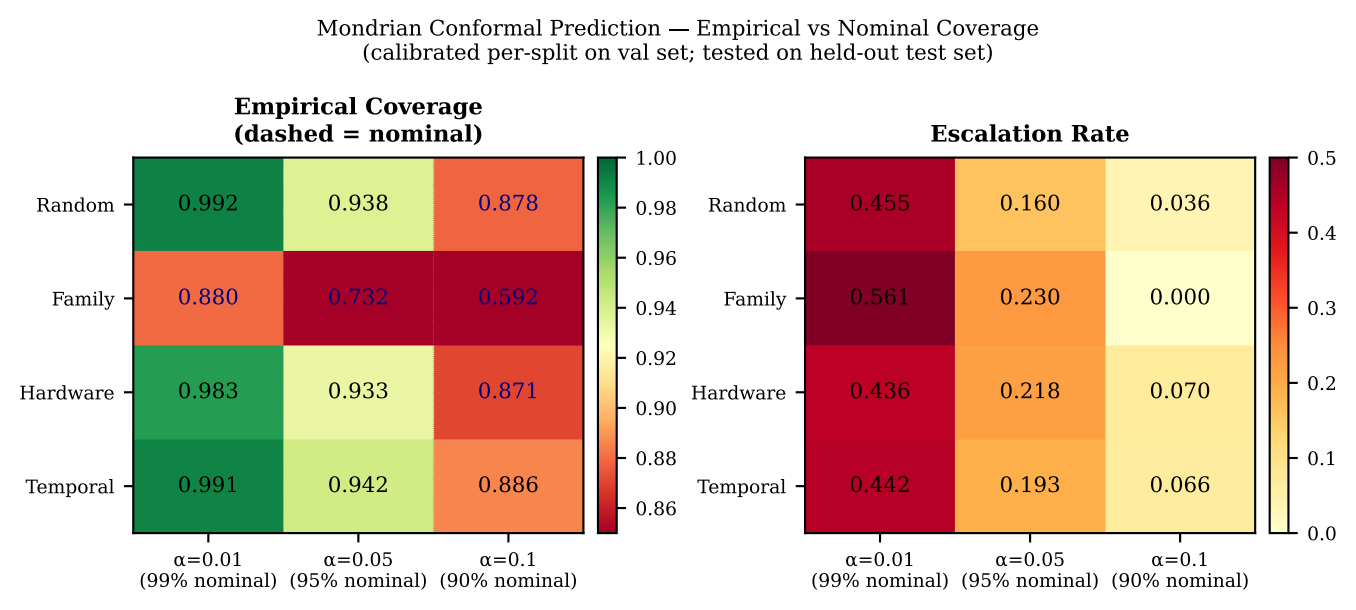
Mondrian Conformal Prediction — Empirical vs Nominal Coverage
(calibrated per-split on val set; tested on held-out test set)
Empirical Coverage
(dashed = nominal)
Random
Family
Hardware
Temporal
0.992
0.938
0.878
0.880
0.732
0.592
0.983
0.933
0.871
0.991
0.942
0.886
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)
Escalation Rate
Random
Family
Hardware
Temporal
0.455
0.160
0.036
0.561
0.230
0.000
0.436
0.218
0.070
0.442
0.193
0.066
0.5
0.4
0.3
0.2
0.1
0.0
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)